Adarsh Singh Tomar

Machine Learning Intern | Deep Learning & NLP | RAG Researcher

adarshsinghtomar7909043383@gmail.com   9508557310

Electronic City Phase 1, Bangalore 560100   19/09/2004   Github   Linkedin

Profile

Machine Learning enthusiast with hands-on experience in Python, TensorFlow, Scikit-learn, and Deep Learning. Skilled in building real-world AI applications including RAG pipelines, NLP models, and MLOps systems. Strong foundation in data science and model deployment. Actively seeking ML research or engineering internships.

Professional Experience

WNS Global Services, *Research Intern – Generative AI & RAG* — 03/2025 – 06/2025, Bangalore, India

- Researched and implemented Retrieval-Augmented Generation (RAG) techniques: **Vector RAG**, **Corrective RAG**, and **Graph RAG**
- Integrated **AWS Neptune**, **Neo4j**, and **OpenSearch** for knowledge graph-based retrieval pipelines
- Built and deployed interactive Q&A apps using **LangChain**, **Groq**, **Streamlit**, **SPARQL**, and **Gremlin**
- Explored and tested the capabilities of **Vibe Coding (Cursor-based AI Coding Assistant)** by building structured apps
- Analyzed Vibe Coding's **benefits, limitations, and comparison with low-code/no-code tools** through multiple real-world prototypes
- Documented research insights for team handoff and internal knowledge sharing

Education

Scaler School of Technology / BITS Pilani, — 07/2023 – 07/2027
Bachelor of Science in Computer Science — Bangalore

Skills

Machine Learning
- Supervised & Unsupervised Learning
- Classification, Regression, Clustering
- Neural Networks, CNNs, RNNs, LSTM
- Transfer Learning, Model Optimization

Data Science
- Data Cleaning, Feature Engineering, EDA
- Data Visualization

Retrieval-Augmented Generation (RAG)
- Vector RAG, Corrective RAG, Graph RAG, LangChain, FAISS, Groq, SPARQL, Gremlin

Programming Language
- Python
- Java
- Spring Boot (Beginner) – REST APIs, DI, Annotations
- SQL, MongoDB

Tools & Platforms
- Git, VS Code, Jupyter Notebook

Deep Learning
- Tokenization, TF-IDF, Word Embeddings
- Transformers, Sentiment Analysis, Text Classification

Natural Language Processing (NLP)
- MLflow, FastAPI, LangChain, Streamlit
- Flask, Docker

MLOps
- MLflow, FastAPI, LangChain, Streamlit
- Flask, Docker

Knowledge Graphs & Retrieval
- Neo4j, AWS Neptune, OpenSearch
- SPARQL, Gremlin, FAISS

Projects

Next-Word Prediction Using LSTM, *Python, TensorFlow, Keras, NLP, Streamlit*
- Built an LSTM-based next-word predictor on Shakespeare's *Hamlet*; deployed via Streamlit

Sentiment Analysis – IMDB Reviews, *Python, Keras, TensorFlow*
- Developed RNN classifier achieving high accuracy in movie review sentiment

Sentiment Analysis – Customer Reviews, *Python, Scikit-learn, TensorFlow, Flask*
- Preprocessed reviews (TF-IDF, stop-word removal); trained Naïve Bayes, LSTM, Transformer; deployed via Flask

End-to-End MLOps Pipeline, *Python, MLflow, FastAPI*
- Created automated model training, versioning, and REST API deployment pipeline using MLflow